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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

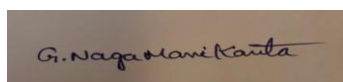
Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **ATHIVARAM JAYAPRAKASH (192425311)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

MANIKANTA

Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						

Total						
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QUESTIONS:

01

Total Marks:25

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

5. Proposed Methodology

Explain the overall methodology adopted in the article. Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →

Classification/Decoder → Output

Identify the role of the major components in the architecture.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score , Loss , Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting

Limited generalization, Model complexity.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Title:Classification of Alzheimer Disease Using DenseNet-201 Based on Deep Transfer Learning Technique

Authors:Zia-ur-Rehman, Mohd Khalid Awang, Javed Rashid, Ghulam Ali, Muhammad Hamid, Samy F. Mahmoud, Dalia I. Saleh, Hafiz Ishfaq Ahmad

Journal:PLOS ONE, Volume 19, Issue 9, Article e0304995

Year of publication: 2024

DOI:<https://doi.org/10.1371/journal.pone.0304995>

Topic / DL technique addressed: Transfer Learning using a pretrained DenseNet-201 convolutional neural network for multi-class medical image classification

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

Response:The article addresses automatic classification of Alzheimer's Disease (AD) stages from brain MRI scans.

AD is a progressive, currently incurable neurodegenerative disease, so identifying how far it has advanced — from no dementia to severe dementia — matters for planning timely care, and manual radiological reading is slow and depends on the reader's experience.

The authors point out that most earlier deep-learning studies on this problem classify only four stages (Non-Demented, Mild, Very Mild and Moderate Demented) and leave out Severe Demented cases altogether, and that comparable transfer-learning approaches in the literature reach roughly 78–92% accuracy or precision even on that easier four-class version of the task. This gap in stage coverage and performance is what the proposed DenseNet-201 model is built to close.

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve? **Response:**The study has three main objectives.

- Build a transfer-learning model based on DenseNet-201 that classifies brain MRI scans into five AD stages (Non-Demented, Mild, Very Mild, Moderate and Severe Demented), instead of the four stages most earlier work used.

- Measure how much data augmentation improves accuracy and generalization compared with training on the raw dataset alone.
- Benchmark the proposed model's accuracy against existing state-of-the-art ADclassification studies.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets? **Response:**The article uses a single MRI dataset, described below.

- Dataset: “AD5C”, a brain-MRI dataset compiled from multiple hospitals in Lahore, Pakistan, and shared publicly via Kaggle.
- Size: 2,380 MRI images in total, across five classes — Mild Demented (372), Very Mild Demented (357), Moderate Demented (637), Non-Demented (324) and Severe Demented (691).
- Data type: 2-D structural MRI brain-scan images in JPG format; most images are 176×208 pixels, with some size variation across the set.
- Split: divided into training, validation and test subsets, with final accuracy reported on the held-out test portion (unseen data) rather than on data used for training.

5. Proposed Methodology

Explain the overall methodology adopted in the article. Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

Response:The overall approach is transfer learning rather than training a network from scratch.

Instead of starting from randomly initialized weights, the authors begin with a DenseNet-201 model whose weights already encode general image features, and adapt it to the five-class AD-staging task — a sensible choice, since training a 201-layer network from scratch on only a few thousand images would risk severe overfitting.

The deep learning architecture used is DenseNet-201 (a 201-layer densely connected convolutional network), used here as the feature extractor for the MRI images. Its key components are the dense blocks and transition layers that extract and progressively compress visual features, followed by a pooling step and a newly added, task-specific classification layer trained to output one of the five AD stages.

For preprocessing, MRI images were resized/standardized to a common input size; in one of the two reported experiments, data-augmentation transformations were additionally applied to the training images to enlarge and diversify the training set.

For training, the network was run with a batch size of 32 for 50 epochs, with training and validation accuracy/loss tracked at every epoch. The authors repeat this procedure twice —

once without data augmentation and once with it — so its effect on performance could be measured directly.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

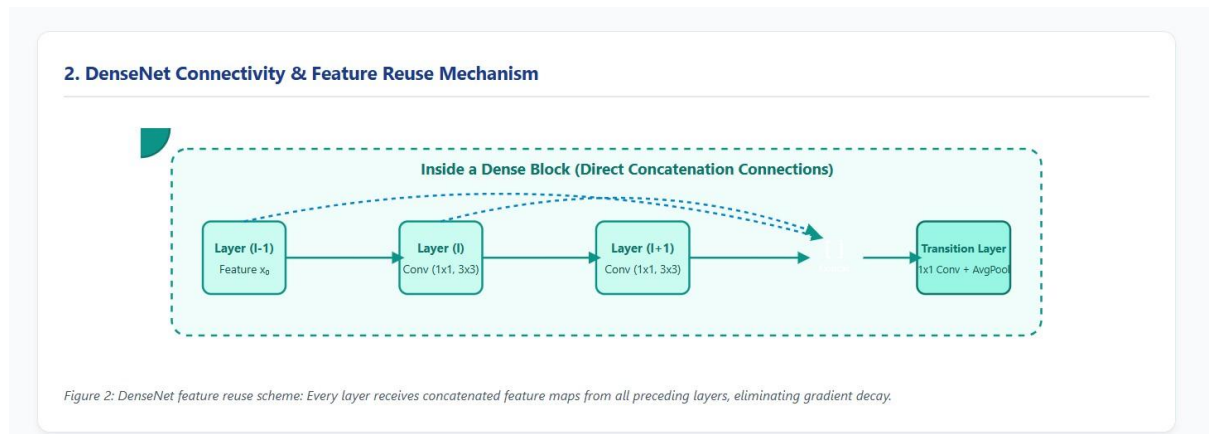
How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Response: The article's central technique is transfer learning applied to a DenseNet-201 backbone.



Rather than learning visual features for MRI images from zero, the model reuses filters DenseNet-201 already learned on a large general-purpose image collection, and only the final classification layers are trained specifically for the AD-staging task. This shortens training time and reduces the amount of labelled medical data needed — a real advantage here, since only 2,380 labelled MRI images were available.

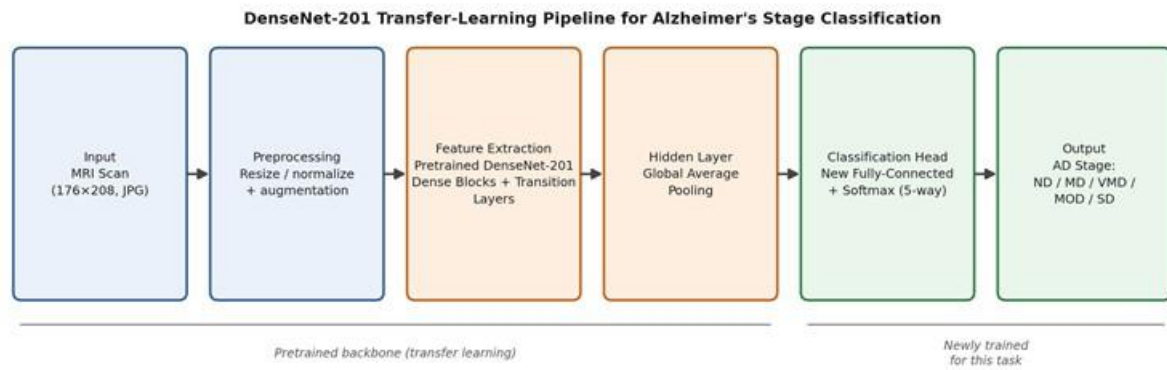
DenseNet performs feature reuse through its dense-connectivity pattern: inside each dense block, every layer receives the concatenated feature maps of all preceding layers in that block, not just the output of the immediately previous layer as in a standard CNN. Later layers can therefore directly reuse low- and mid-level features learned earlier in the network. This improves gradient flow during backpropagation (helping a network this deep train stably), encourages more compact, non-redundant feature learning, and is a major reason DenseNet-201 reaches high accuracy without an excessive parameter count.

The remaining sub-questions in this section (on RNN/LSTM, Bidirectional RNN, Encoder–Decoder/Seq2Seq and BPTT) do not apply, since the selected article's technique is transfer learning with DenseNet rather than a recurrent architecture.

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Response: A simplified version of the model's data flow, redrawn from the transfer-learning building block shown in the article, is given below.



1. Overall Research Methodology Flowchart



End-to-End Transfer Learning Strategy (Training for 50 Epochs, Batch Size = 32)

Figure 1: High-level methodology pipeline showing transfer learning from MRI input to 5-stage classification.

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers → Classification/Decoder → Output

Identify the role of the major components in the architecture.

Explanation: Input MRI images are resized and preprocessed (and, in the augmented run, transformed) before entering the pretrained DenseNet-201 backbone, whose dense blocks and transition layers extract increasingly abstract visual features from the scan. A global-average pooling layer condenses these deep feature maps into a single feature vector, which feeds a newly trained fully connected layer with a 5-way softmax output — one probability per AD stage (ND, MD, VMD, MOD, SD). The pretrained convolutional layers act as a general purpose visual feature extractor (transfer learning), while only the pooling-to-output portion of the network is learned specifically for this task.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score, Loss, Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

Response: The article reports a consistent, clearly-defined set of evaluation metrics.

- Metrics used: Accuracy, Precision, Recall, F1-score and ROC/AUC (training/validation Loss is also tracked as a secondary diagnostic).
- Without data augmentation: 96.92% accuracy on the held-out test set.
- With data augmentation: 98.24% accuracy and an overall AUC of 97.36%.

- Per-class ROC-AUC with augmentation: Very Mild 100%, Moderate 99%, Severe 99%, Non-Demented 98%, Mild 94% — consistently strong across all five classes, including the previously under-studied Severe Demented class.

9. Results and Discussion

What are the major findings of the research?

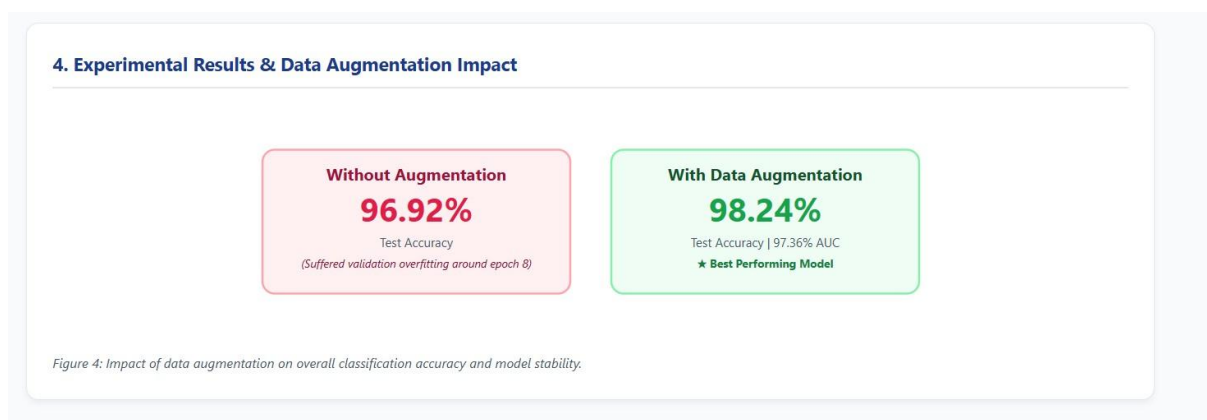
How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

Response: The clearest finding is that data augmentation meaningfully improved both accuracy and generalization.

Without augmentation, training accuracy kept climbing toward 98% through 50 epochs while validation accuracy peaked near 88% around epoch 8 and then fell below 80% by epoch 50, with validation loss rising after epoch 3 — a classic overfitting pattern on a fairly small dataset. Augmentation (98.24% accuracy) narrowed this gap and produced steadier validation performance than the non-augmented run (96.92%).



Compared with existing methods, the proposed model performs favourably: the article cites a connection-wise-attention CNN ensemble reaching about 88.9% accuracy on the ADNI dataset, an AlexNet-based transfer-learning model (“ADDTLA”) reaching about 91.7% precision on a four-class version of the problem, and a VGG19-based model reaching about 77.7% accuracy on four classes. The DenseNet-201 model here exceeds all three while additionally handling a harder, five-class version of the task that includes Severe Demented cases.

The best results come from DenseNet-201 with data augmentation. The main contributing factors appear to be DenseNet's dense connectivity (efficient feature reuse and stable gradients even at 201 layers), transfer learning (a head start from general visual features rather than learning them from a small medical dataset), and augmentation (reduced overfitting from exposure to more varied versions of the limited training images).

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach. **Response:** At least three advantages stand out.

- Strong performance across all five AD stages, including the Severe Demented class that most earlier studies leave out entirely, without sacrificing accuracy on the other four.

- Transfer learning greatly reduces the labelled data and training time needed to get a very deep (201-layer) network working well — important in medical imaging, where large labelled datasets are hard to obtain.
- DenseNet's dense connectivity gives efficient feature reuse and strong gradient flow, so the model reaches high accuracy (98.24%) and a high AUC (97.36%) without the training instability very deep plain CNNs can suffer from.
- The paper directly demonstrates, through its two experiments, how much data augmentation improves robustness — a practical, reusable insight for similar small-dataset medical-imaging problems.

3. AD5C Dataset Class Distribution (2,380 Total Images)



Figure 3: Sample distribution across 5 AD stages (*Severe Demented is included here, unlike conventional 4-stage studies).

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

Response: Several limitations are worth noting, some flagged by the authors and some observed independently.

- Overfitting without augmentation: the gap between rising training accuracy and falling validation accuracy in the non-augmented run shows the base model is quite sensitive to the amount of training data available.
- Dataset scale and diversity: 2,380 images from hospitals in a single city (Lahore) is modest by deep-learning standards, and some classes have fewer than 400 images; the article does not test the model on scans from other hospitals, scanners or populations.
- Class imbalance: Non-Demented (324) and Severe Demented (691) differ by roughly 2:1, which can bias a classifier toward better-represented classes even when aggregate metrics look strong.
- Computational cost: DenseNet-201 is a deep, 201-layer network, heavier to train and deploy than shallower CNNs — a practical constraint for real clinical settings that the article does not directly address.
- 2-D, slice-level classification: the model works on individual 2-D MRI images rather than full 3-D scan volumes, which may discard spatial context a volumetric approach could use.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance? **Response:** Working through the five prompts for this section:

Is the architecture appropriate? Yes — DenseNet-201 combined with transfer learning is a well-justified choice: it is a proven architecture for image classification, and starting from pretrained weights directly addresses the small-dataset constraint typical of medical imaging. Is the dataset adequate? It is adequate for a first demonstration of five-class AD staging, but modest in scale and drawn from a single geographic source, so it is not yet adequate for claims of broad clinical generalizability.

Are the evaluation metrics appropriate? Yes — accuracy, precision, recall, F1 and AUC together cover both overall correctness and per-class discriminative performance, and the perclass ROC values are especially useful given the dataset's class imbalance.

Are the results convincing? The augmented-model results are strong, and the authors' own reporting of the non-augmented model's overfitting is a reassuring sign of transparent reporting rather than cherry-picking; even so, without an external, independently sourced test set, 98.24% is best read as a strong internal benchmark rather than a settled clinical-grade result.

Could another Unit IV technique perform better? Possibly — a sequence-aware technique, such as treating a stack of MRI slices as an ordered sequence and applying an RNN/LSTM or an attention-based encoder–decoder across slices, could capture inter-slice spatial continuity that single-slice DenseNet-201 classification discards, and would be a reasonable technique to benchmark against in future work.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Response: Building on the limitations above, several directions could extend this work.

- Validate the model on larger, multi-hospital, multi-scanner datasets to test generalization beyond the AD5C dataset.
- Add explainability tools such as Grad-CAM or attention maps so clinicians can see which brain regions influenced a given prediction.
- Extend the approach from 2-D slice classification to full 3-D volumetric MRI analysis.
- Explore lighter-weight or compressed versions of the architecture for deployment on limited clinical hardware.
- Combine MRI features with non-imaging clinical or cognitive-test data (multimodal fusion) for a more complete diagnostic picture.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

Response:In summary, the article tackles automatic staging of Alzheimer's Disease from MRI scans, proposing a DenseNet-201 transfer-learning model that, unlike most prior work, covers five stages including Severe Demented.

Trained and evaluated on the 2,380-image AD5C dataset, the model reaches 96.92% accuracy without data augmentation and 98.24% accuracy (97.36% AUC) with it, outperforming the comparison methods cited in the article. Overall, this is a solid, clearly evidenced contribution whose main limitations are the moderate size and single-site origin of the dataset and the absence of external validation — both reasonable directions for follow-up work rather than flaws that undermine the core result.

15. References

Provide the complete citation of the selected research article and any additional references used in the review. **Response:**

Zia-ur-Rehman, Awang, M. K., Rashid, J., Ali, G., Hamid, M., Mahmoud, S. F., Saleh, D. I., & Ahmad, H. I. (2024). Classification of Alzheimer disease using DenseNet-201 based on deep transfer learning technique. PLOS ONE, 19(9), e0304995.
<https://doi.org/10.1371/journal.pone.0304995>

Note: the comparison studies discussed in Sections 2 and 9 (the connection-wise-attention CNN ensemble on ADNI, the AlexNet-based ADDTLA model, and the VGG16/VGG19-based model) are cited as related work within the article above; see its reference list for their full bibliographic details.